Vector Spaces I–II: Subspaces, Span, Linear Independence, Basis and Dimension

From the class book: Chapter 3, Sections 3.1 to 3.4

1. Key definitions

Vector space. A vector space over a field F is a set V with addition and scalar multiplication such that $(V, +)$ is an abelian group and, for all $a, b \in F$, $u, v \in V$: $a(u+v) = au + av$, $(a+b)v = av + bv$, $(ab)v = a(bv)$, $1v = v$.

- Consequences: $0v = 0$, $a0 = 0$, $(-1)v = -v$, and $av = 0$ forces $a = 0$ or $v = 0$.
- Examples: F^n, $M_{m\times n}(F)$, $F[x]$, P_n (degree $\le n$), $C[a,b]$, all real sequences.
- If $K \subseteq L$ are fields, L is a vector space over K: $\dim_{\mathbb{R}} \mathbb{C} = 2$, $\dim_{\mathbb{Q}} \mathbb{Q}(\sqrt{2}) = 2$, while $\mathbb{R}$ and $\mathbb{C}$ are infinite dimensional over $\mathbb{Q}$.
- $\mathbb{R}$ is **not** a vector space over $\mathbb{C}$ ($i \cdot 1 \notin \mathbb{R}$); polynomials of degree exactly n do not form a vector space (no 0).

Subspace. A non-empty $W \subseteq V$ is a subspace iff $au + bv \in W$ for all $u, v \in W$, $a, b \in F$ (one-step test). Every subspace contains 0.

Span, independence, basis. $\operatorname{span} S$ is the set of finite linear combinations of S (the smallest subspace containing S; $\operatorname{span} \varnothing = \{0\}$). S is **linearly independent** if $a_1 v_1 + \cdots + a_k v_k = 0$ with distinct $v_i \in S$ forces all $a_i = 0$. A **basis** is an independent spanning set; all bases have the same size, $\dim V$.

Space	over $\mathbb{C}$	over $\mathbb{R}$	over $\mathbb{Q}$
$\mathbb{C}^n$	n	$2n$	∞
$\mathbb{R}^n$	not a VS	n	∞
$\mathbb{Q}(\sqrt{2})^n$	not a VS	not a VS	$2n$
$M_{m\times n}(\mathbb{C})$	mn	$2mn$	∞

2. Main results

Subspace or not?

- **Yes:** sets cut out by homogeneous linear conditions, such as $\{x + y = 0\}$, $\{\operatorname{tr} A = 0\}$, symmetric, skew-symmetric or upper triangular matrices, $\{X : XA = AX\}$, $\{f : f(1/2) = 0\}$, solutions of $y'' + py' + qy = 0$, convergent sequences.
- **No:** a non-zero constant term ($x + y + z = 1$, solutions of $y'' + y = \sin x$); non-linear conditions ($xy = 0$, $\det A = 0$, $\operatorname{rank} A = 1$, $xy \geq 0$, invertible matrices); $\mathbb{Q}^3$ inside $\mathbb{R}^3(\mathbb{R})$.
- Watch for disguised linear conditions: over $\mathbb{R}$, $x^2 + y^2 = 0$ means $x = y = 0$, a line.

Theorem (Operations on subspaces)

- Any intersection of subspaces is a subspace.
- $W_1 \cup W_2$ is a subspace iff $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.
- $W_1 + W_2$ is the smallest subspace containing $W_1 \cup W_2$.
- Over an infinite field, V is never a finite union of proper subspaces. Over $\mathbb{F}_q$, $\mathbb{F}_q^2$ is the union of its $q + 1$ lines, and no fewer proper subspaces suffice.

Important results: linear independence

- A set containing 0 is dependent; subsets of independent sets are independent; supersets of dependent sets are dependent.
- If S is independent and $v \notin \operatorname{span} S$, then $S \cup \{v\}$ is independent.
- n vectors in F^n are independent iff the matrix with these columns has non-zero determinant.
- If $2 \neq 0$: $\{u, v\}$ independent $\iff$ $\{u + v, u - v\}$ independent; $\{u, v, w\}$ independent $\iff$ $\{u + v, v + w, w + u\}$ independent.
- With $u_1, \ldots, u_n$ independent and $v_i = u_i + u_{i+1}$ (mod n): $v_1, \ldots, v_n$ are independent iff n is odd (for even n they span $n - 1$ dimensions). With $v_i = u_i - u_{i+1}$ they are always dependent.
- Non-zero polynomials of distinct degrees are independent; so are $e^{r_1 t}, \ldots, e^{r_k t}$ for distinct r_i, and $1, \cos t, \sin t, \cos 2t, \sin 2t, \ldots$.
- A Wronskian non-zero at one point gives independence; the converse fails (t^2 and $t|t|$).

Theorem (Dimension theorems, $\dim V = n$)

- An independent set has at most n vectors; a spanning set has at least n. An independent set or a spanning set with exactly n vectors is a basis.
- Every independent set extends to a basis; every spanning set contains a basis.
- A subspace W has $\dim W \leq n$, with equality iff $W = V$.
- A subspace of F^n cut out by k independent homogeneous linear equations has dimension $n - k$.

3. Standard dimensions

Space	Dimension
P_n; polynomials of degree $\leq n$ in 2; in 3 variables	$n+1$; $\binom{n+2}{2}$; $\binom{n+3}{3}$
homogeneous polynomials of degree d in k variables	$\binom{d+k-1}{k-1}$
$M_{m\times n}(F)$; symmetric; skew-symmetric	mn; $\frac{n(n+1)}{2}$; $\frac{n(n-1)}{2}$
upper triangular; diagonal; trace zero	$\frac{n(n+1)}{2}$; n; n^2-1
symmetric with trace 0	$\frac{n(n+1)}{2} - 1$
Hermitian (or skew-Hermitian) $n \times n$, over $\mathbb{R}$	n^2
all row and column sums 0	$(n-1)^2$
$\{X : XA = AX\}$, A with n distinct eigenvalues	n; for $A = \mathbf{diag}$ with blocks of sizes n_i: $\sum n_i^2$
solutions of a linear homogeneous ODE of order n	n
$\mathbf{span}\{I, B, B^2, \ldots\}$	degree of the minimal polynomial of B

Method: dimension of a subspace given by conditions.

- (1) Write each condition as a homogeneous linear equation in the coordinates.
- (2) Count the independent equations k: conditions on disjoint sets of variables are automatically independent; otherwise row reduce.
- (3) The dimension is (number of coordinates) $-\ k$. Check with a small explicit basis when possible.

Method: testing independence.

- Vectors in F^n: put them as columns and compute the rank (or the determinant when there are n of them).
- Polynomials or matrices: use coordinates in the standard basis.
- Functions: evaluate at well-chosen points, or use a known identity ($\cos^2 + \sin^2 = 1$, addition formulas) to exhibit a dependence.

4. Exam shortcuts and common traps

- The first check for "subspace" is $0 \in W$; the second is closure under sums (most non-linear sets fail here).
- Symmetric polynomials in x, y are polynomials in $x+y$ and xy; of degree $\leq d$ they have a basis $(x+y)^a(xy)^b$, $a+2b \leq d$.
- For $(1,1,x),(1,x,1),(x,1,1)$: $\det = -(x-1)^2(x+2)$, a basis iff $x \neq 1, -2$.
- If $B^2 = I$, $B \neq \pm I$, every polynomial in B is $aI + bB$: dimension 2.
- **Traps:** $\mathbb{R}$ over $\mathbb{C}$ is not a vector space; the union of two subspaces is usually not a subspace; a zero Wronskian does not prove dependence; "dimension of $\mathbb{Q}(\sqrt{2})^2$" depends on the field (4 over $\mathbb{Q}$, 2 over $\mathbb{Q}(\sqrt{2})$).

5. Solved examples

Example 1. Find $\dim V$ for $V = \{x \in \mathbb{R}^{60} : x_1 = \cdots = x_{20},\ x_{21} + \cdots + x_{60} = 0,\ x_1 = x_{60}\}$.

Solution

- 19 equations $x_1 = x_i$ ($2 \leq i \leq 20$), one sum condition, and $x_1 = x_{60}$.
- They are independent: the last links two blocks that were otherwise free, so it is not a consequence of the others.
- $\dim V = 60 - 21 = 39$.

Example 2. Find the dimension of the real 5×5 matrices with all row sums and column sums 0.

Solution

- Choose the leading 4×4 block freely (16 parameters).
- The last entry of each row and of each column is then forced; the corner entry is consistent, since both ways it equals the sum of the block.
- So the dimension is $(5-1)^2 = 16$.

Example 3. For which real x are $(1,1,x),(1,x,1),(x,1,1)$ a basis of $\mathbb{R}^3$?

Solution

- Add all columns to the first: each row sums to $x+2$, so $\det = (x+2)\det\begin{pmatrix} 1 & 1 & x \\ 1 & x & 1 \\ 1 & 1 & 1 \end{pmatrix}$.
- Subtract row 3 from rows 1 and 2: the determinant becomes $(x+2)\cdot(-(x-1)^2)$.
- So the vectors form a basis iff $x \neq 1$ and $x \neq -2$.

Example 4. Let $u_1, \dots, u_6$ be independent and $v_i = u_i + u_{i+1}$ (mod 6). Find $\dim \operatorname{span}\{v_i\}$.

Solution

- $v_1 - v_2 + v_3 - v_4 + v_5 - v_6 = 0$, so the v_i are dependent.
- $v_1, \dots, v_5$ are independent: their coefficient matrix restricted to $u_1, \dots, u_5$ is bidiagonal with 1s on the diagonal.
- So the dimension is 5.

Example 5. Find $\dim\{p \in \mathbb{R}[x, y] : \deg p \le 4,\ p(x, y) = p(y, x)\}$.

Solution

- Basis $(x + y)^a (xy)^b$ with $a + 2b \le 4$.
- $b = 0$: $a = 0, \dots, 4$ (5); $b = 1$: $a = 0, 1, 2$ (3); $b = 2$: $a = 0$ (1).
- Dimension 9.

Example 6. Decide which are independent: $\{1, \cos^2 t, \sin^2 t\}$, $\{t^2, t|t|\}$, $\{\sin t, \cos t, \sin(t + 1)\}$.

Solution

- $\cos^2 t + \sin^2 t - 1 = 0$: dependent.
- $at^2 + bt|t| = 0$ at $t = 1$ and $t = -1$ gives $a + b = 0$ and $a - b = 0$: independent, although the Wronskian is 0 everywhere.
- $\sin(t + 1) = \cos 1 \ \sin t + \sin 1 \ \cos t$: dependent.

Example 7. What is the least number of proper subspaces whose union is $\mathbb{F}_3^2$? And $\mathbb{R}^2$?

Solution

- Each proper non-zero subspace of $\mathbb{F}_3^2$ is a line with 2 non-zero vectors; there are 8 non-zero vectors, so at least 4 lines are needed, and the 4 lines through 0 suffice.
- $\mathbb{R}^2$ is never a finite union of proper subspaces (the field is infinite).